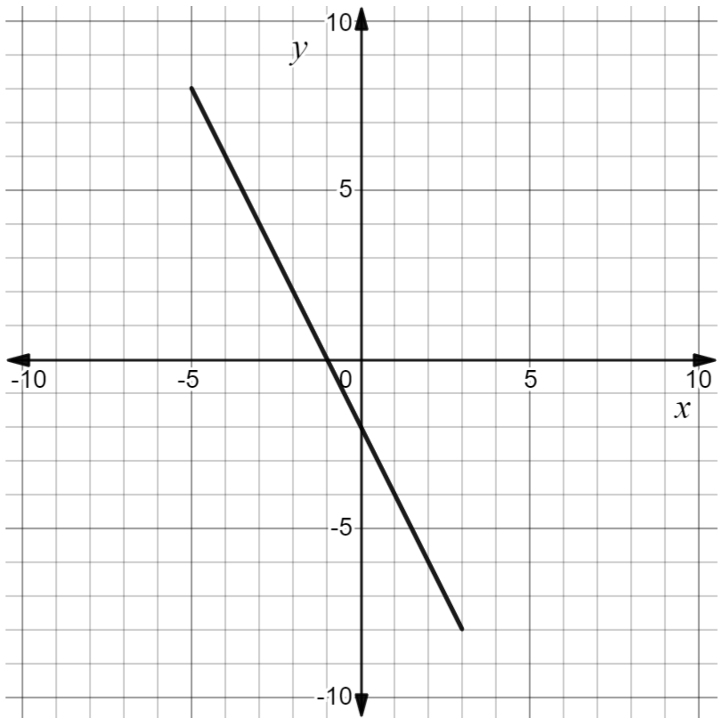
Algebra 1
Unit 4
Lesson 1 Practice

Name Answer Key______________
Date______________________
Period_____________________

The graph of $y = -\frac{7}{3}x - 2$ is shown. The graph has a restricted domain.




1. Give a written description for the range.
   All values greater than or equal to $-8$ and less than or equal to $8$.

2. Write the domain using inequalities.
   $-5 \leq x \leq 3$

3. Write the domain using set-builder notation.
   $\{x: -5 \leq x \leq 3\}$

4. Write the range using set-builder notation.
   $\{y: -8 \leq y \leq 8\}$

For each written description complete the table.

| Written Description | Inequalities | Set-builder notation |
|---|---|---|
| 5. All values less than or equal to $5\frac{3}{4}$ | $x \leq 5\frac{3}{4}$ | $\left\{x: x \leq 5\frac{3}{4}\right\}$ |
| 6. All values greater than $-2.6$ but less than or equal to $8.9$ | $-2.6 < x \leq 8.9$ | $\{x: -2.6 < x \leq 8.9\}$ |
| 7. All values greater than $-\frac{22}{6}$ | $x > -\frac{22}{6}$ | $\left\{x: x > -\frac{22}{6}\right\}$ |

8. Write a written description of the set of values represented by $\{x: 30.2 > x \geq -3.8\}$.
All values greater than or equal to $-3.8$ but less than $30.2$.

9. Write the set of values represented by $\{h \mid h < -0.25\}$ using inequalities.
$h < -0.25$ *or* $-0.25 > h$

10. Write the set of values represented by $c \geq \frac{5}{6}$ using set-builder notation.
$\left\{c: c \geq \frac{5}{6}\right\}$

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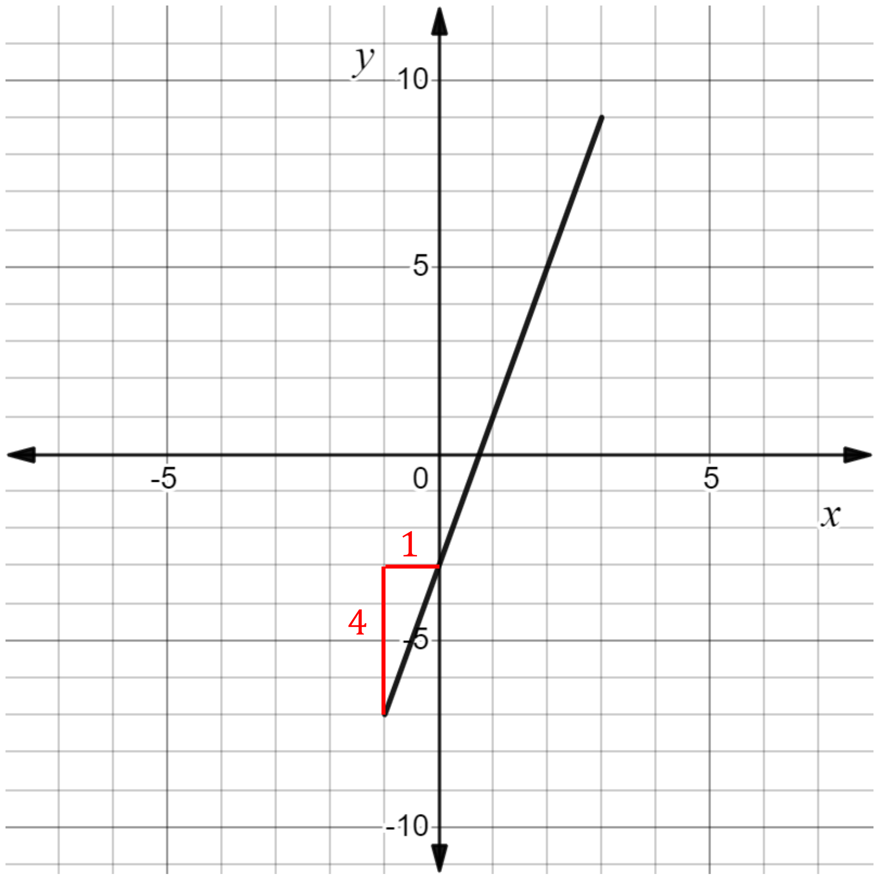
Lesson 2 Practice Period_____________________

The graph of linear function is shown. The graph has a restricted domain.




1. The slope of the line segment is $\frac{4}{1} = 4$.

2. As the $x$-values are increasing, the $y$-values are increasing.

3. The coordinates of the point where the line crosses the $y$-axis are $(0, -3)$.
The $y$-value of the point is $-3$.

4. The coordinates of the point where the line crosses the $x$-axis are $\left(\frac{3}{4}, 0\right)$.

The $x$-value of the point is $\frac{3}{4}$.

Identify the key features of the horizontal line $y = -3$ in questions 5-6.

5.

| Domain | Range | As $x$-values increase, $y$-values ... |
|---|---|---|
| $\{x: -\infty < x < \infty\}$ | $\{y: y = -3\}$ | Stay the same |

6.

| Slope | $x$-intercept | $y$-intercept |
|---|---|---|
| $0$ | none | $(0, -3)$ |

Identify the key features of the line $y = \frac{2}{3}x$ in questions 7-8.

7.

| Domain | Range | As $x$-values increase, $y$-values ... |
|---|---|---|
| $\{x: -\infty < x < \infty\}$ | $\{y: -\infty < y < \infty\}$ | increase |

8.

| Slope | $x$-intercept | $y$-intercept |
|---|---|---|
| $\frac{2}{3}$ | $(0,0)$ | $(0,0)$ |

Identify the key features of the line $y = -2x + 13$ in questions 9-10.

9.

| Domain | Range | As $x$-values increase, $y$-values ... |
|---|---|---|
| $\{x: -\infty < x < \infty\}$ | $\{y: -\infty < y < \infty\}$ | decrease |

10.

| Slope | $x$-intercept | $y$-intercept |
|---|---|---|
| $-2$ | $\left(\frac{13}{2}, 0\right)$ | $(0,13)$ |

Algebra 1 | Name Answer Key
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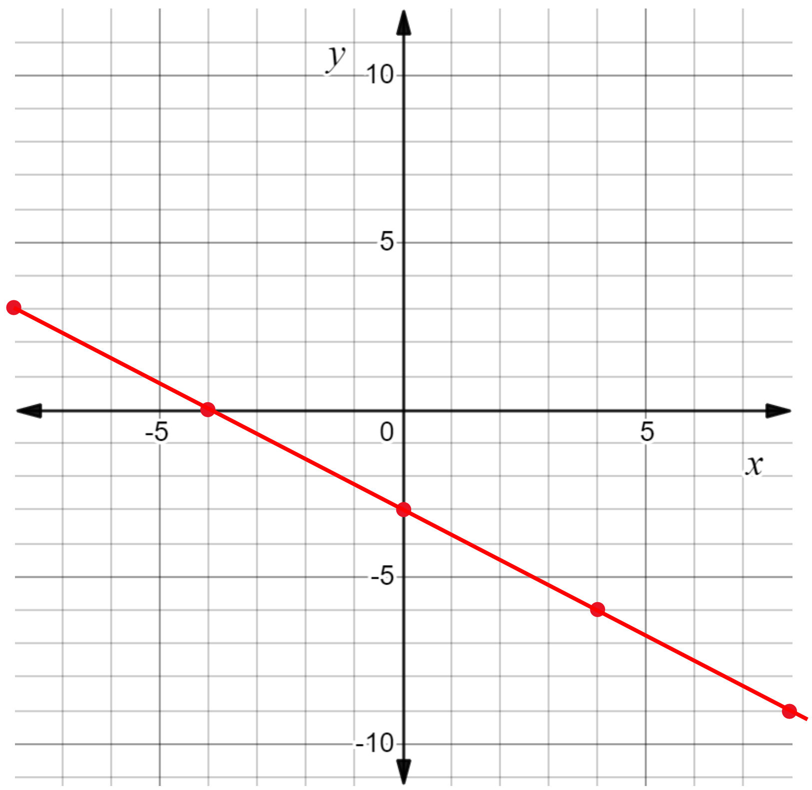
1. Complete the table of values for $y = -\frac{3}{4}x - 3$.

$-\frac{3}{4}(4) - 3 = -3 - 3 = -6$

| $x$ | $-8$ | $-4$ | $0$ | $4$ | $8$ |
|---|---|---|---|---|---|
| $y$ | $3$ | $0$ | $-3$ | $-6$ | $-9$ |

$-\frac{3}{4}(-8) - 3 = 6 - 3 = 3$  $-\frac{3}{4}(-4) - 3 = 3 - 3 = 0$  $-\frac{3}{4}(8) - 3 = -6 - 3 = -9$

2. Graph the equation $y = -\frac{3}{4}x - 3$.




3. Write the domain of the line $y = -\frac{3}{4}x - 3$ using set-builder notation.

$\{x: -\infty < x < \infty\}$

4. What is the $x$-intercept of the line $y = -\frac{3}{4}x - 3$?

$(-4, 0)$

5. Complete the statement about the line $y = -\frac{3}{4}x - 3$.

As $x$-values increase, $y$-values decrease

6. Complete the table of values for $y + 5 = -4(x - 2)$.

$y + 5 = -4(-1 - 2) \rightarrow = y + 5 = -4(-3) \rightarrow y = 12 - 5 = 7$

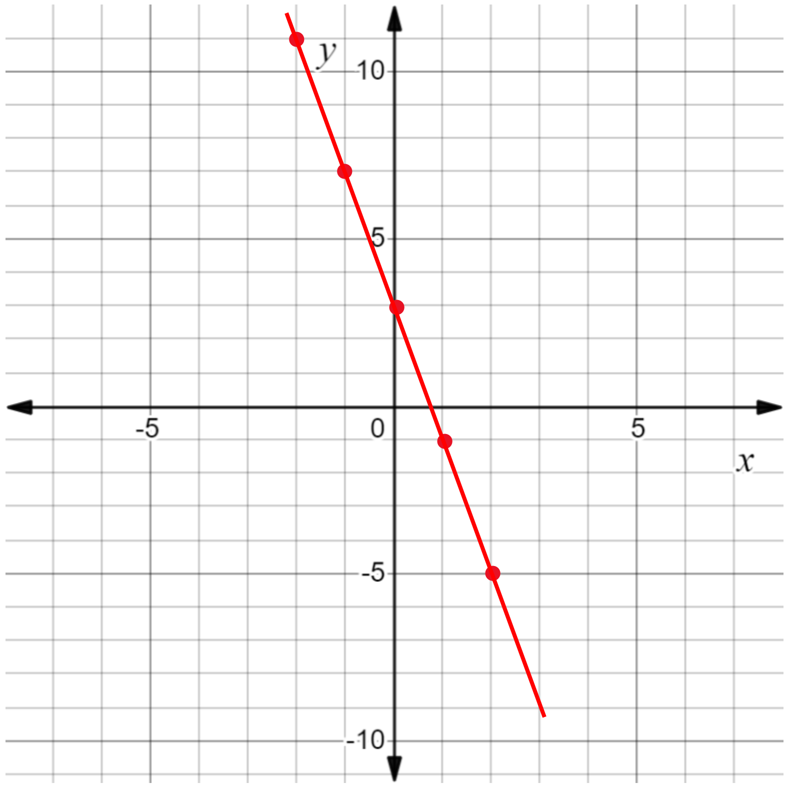
$y + 5 = -4(-1) \rightarrow y = 4 - 5 = -1$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $11$ | $7$ | $3$ | $-1$ | $-5$ |

$y + 5 = -4(-2 - 2) \rightarrow = y + 5 = -4(-4) \rightarrow y = 16 - 5 = 11$

$y + 5 = -4(-2) \rightarrow y = 8 - 5 = 3$

7. Graph the equation $y + 5 = -4(x - 2)$.




8. Use inequalities to complete the table for the line $y + 5 = -4(x - 2)$.

| **Domain** | **Range** |
|---|---|
| $-\infty < x < \infty$ | $-\infty < y < \infty$ |

9. What is the $y$-intercept of the line $y + 5 = -4(x - 2)$?
$(0,3)$

10. Complete the statement about the line $y - 3 = 3(x + 1)$.

As $x$-values increase, $y$-values increase.

Algebra 1 | Name Answer Key
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A study of the average emissions of carbon dioxide ($CO_2$) from a vehicle in the Netherlands was conducted from 2000 to 2019. The function $C(x) = -5x + 190$ measures the carbon intensity of newly registered passenger vehicles in the Netherlands where $x$ is the number of years since 2000 and $C$ is the average emissions of $CO_2$ (in grams) per kilometer.

1. Two students were discussing how to find the domain of the function $C(x) = -5x + 190$. Their explanations are shown.

| Michelle | Jaheim |
|---|---|
| The domain is all real numbers because the function is linear. | The domain should start at $0$ because $x$ represents the number of years since 2000. It would make sense for the domain to end with $19$ because that is the last year of the study. |

Determine which student is correct and write the domain using inequalities.

Jaheim is correct.

$0 \leq x \leq 19$

2. Write the range of the function $C(x) = -5x + 190$ using inequalities.
$95 \leq y \leq 190$

3. Determine the $x$-intercept.
$-5x + 190 = 0$

$-5x = -190$

$x = 38$

$x$-intercept $= 38$

4. Interpret the $x$-intercept.
In the year 2038, the measure of carbon intensity for newly registered passenger vehicles in the Netherlands will reach an average emission rate of $0$ $CO_2$ per kilometer.

5. Determine the $y$-intercept.
$c(0) = -5(0) + 190$
$= 190$

$y$-intercept $= 190$

6. Interpret the $y$-intercept.
At the beginning of the study, in the year 2000, the average carbon emission rate for newly registered passenger vehicles in the Netherlands was measured to be $190$ $CO_2$ per kilometer.

7. Complete the statement about the line $C(x) = -5x + 190$.

As $x$-values increase, $y$-values decrease

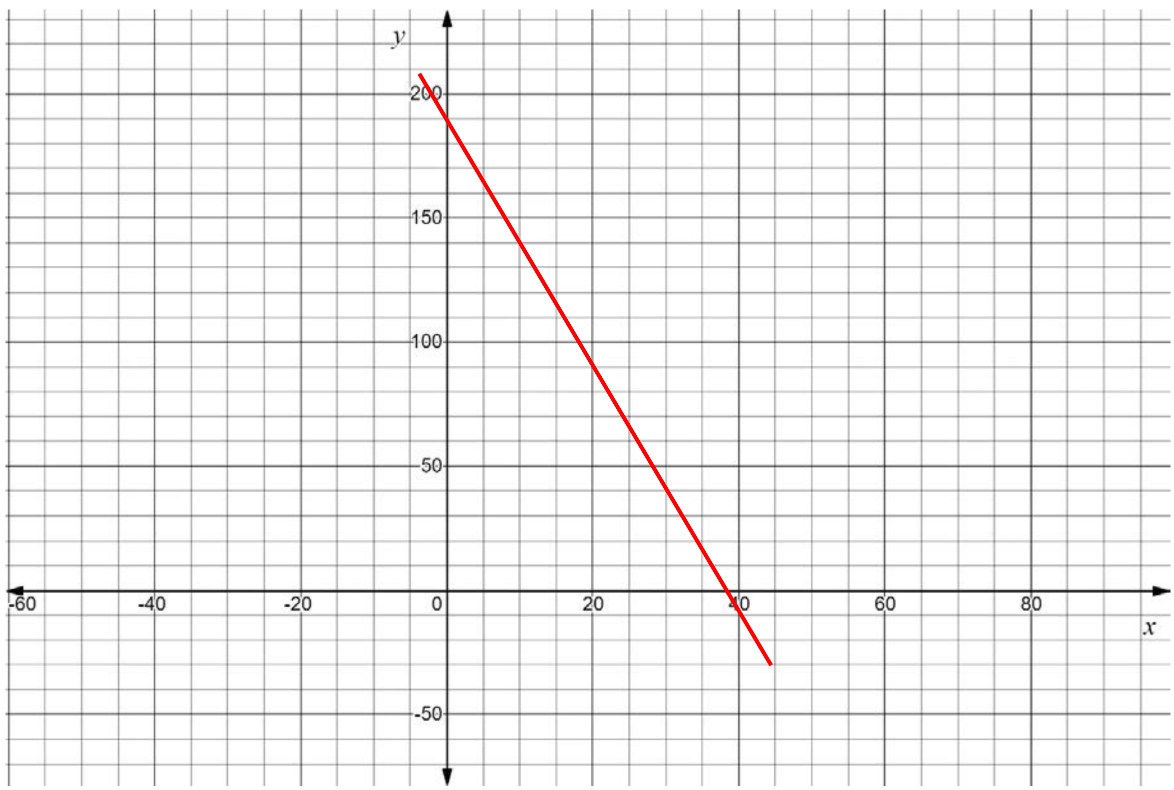
8. Determine the slope.
$y = mx + b$, $m$ represents the slope

$m = -5$

slope $= -5$

9. Interpret the slope.
The average carbon emission rate for newly registered passenger vehicles in the Netherlands decreases by $5$ $CO_2$ per kilogram for each successive year after 2000.

10. Graph the function.



Algebra 1
Unit 4
Lesson 5 Practice

Name Answer Key ___________
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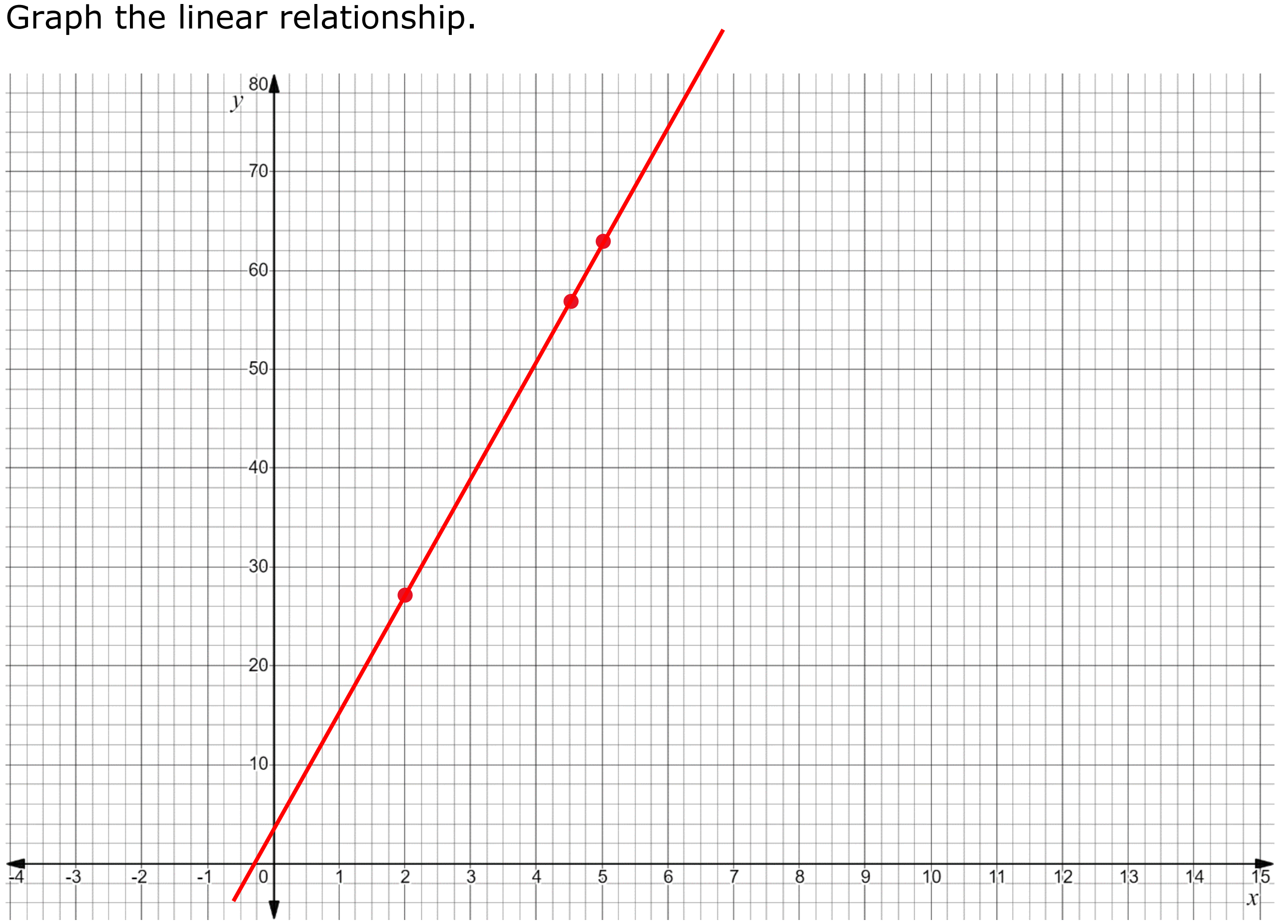
The table of values shows the linear relationship between $x$, the height of water in a container, and $y$, the weight of the container.

| Height of Water (inches) | 2 | 4.5 | 5 | 7.25 | 8 |
|---|---|---|---|---|---|
| Weight of Container (ounces) | 27 | 57 | 63 | 90 | 99 |

1. Graph the linear relationship.




2. Place a check in the box of a key feature that can be determined from the table of values.

| Domain | Range | $x$-intercept | $y$-intercept | Slope | Increasing/ Decreasing |
|---|---|---|---|---|---|
| ✓ | ✓ | | | ✓ | ✓ |

3. The equation $12x - y = -3$ models the linear relationship. Place a check in the box of a key feature that can be determined using the equation.

| **Domain** | **Range** | **$x$-intercept** | **$y$-intercept** | **Slope** | **Increasing/ Decreasing** |
|---|---|---|---|---|---|
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

4. Write the equation $12x - y = -3$ in slope-intercept form.

$-y = -12x - 3$

$y = 12x + 3$

5. What domain makes sense for the context?

$0 \leq x < \infty$ or until the max capacity for the container is reached.

6. What range makes sense for the context?

$3 \leq x < \infty$ or until the max weight for the container is reached.

7. Determine the $x$ –intercept.

$0 = 12x + 3$

$-3 = 12x$

$x = -\frac{1}{4}$

8. Interpret the $x$ –intercept.

When the height of the water is $-\frac{1}{4}$ inches, the container will weigh $0$ oz. (which is impossible- meaning the bucket will always weigh $3$ oz., even without water in it).

9. Determine the slope.

$y = mx + b$

$m = 12$

10. Interpret the slope.

Every inch of water added to the container increases its weight by $12$ oz.

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For each real-world representation of a context write an inequality that represents the constraint.

1. A person has to be at least $18$ years old to vote.
   $p \geq 18$

2. A job offers a salary between $\$42{,}000$ and $\$50{,}000$.
   $42{,}000 < x < 50{,}000$

3. The highest altitude for skydiving without oxygen is $14{,}000$ feet.
   $s \leq 14{,}000$

An automatic soap dispenser holds $18$ ounces (oz) of soap. Each time someone places their hand under the soap dispenser releases $0.15$ oz of soap. The function $f(x) = 18 - 0.15x$ models the amount of soap, in ounces, in the dispenser for $x$ dispenses.

4. Find the value of $x$ for which $f(x) = 13.2$.
   $13.2 = 18 - 0.15x$

   $-18 \quad -18$

   $\frac{-4.8}{-0.15} = -\frac{0.15x}{-0.15}$

   $32 = x$

5. In terms of the context, what does the value of $x$ that is the solution to $f(x) = 13.2$ mean?
   $x(32)$ represents the number of dispenses of soap that are made. So, after $32$ dispenses, there are $13.2$ ounces remaining.

6. Find the $x$ –intercept.
   $f(x) = 18 - 0.15x$

   $0 = 18 - 0.15x$

   $-18 \quad -18$

   $-\frac{18}{-0.15} = -\frac{0.15x}{-0.15}$

   $x = 120$

7. What does the $x$ –intercept mean within the context?
   This means it will take $120$ dispenses of soap to have no (or zero) soap left in the dispenser.

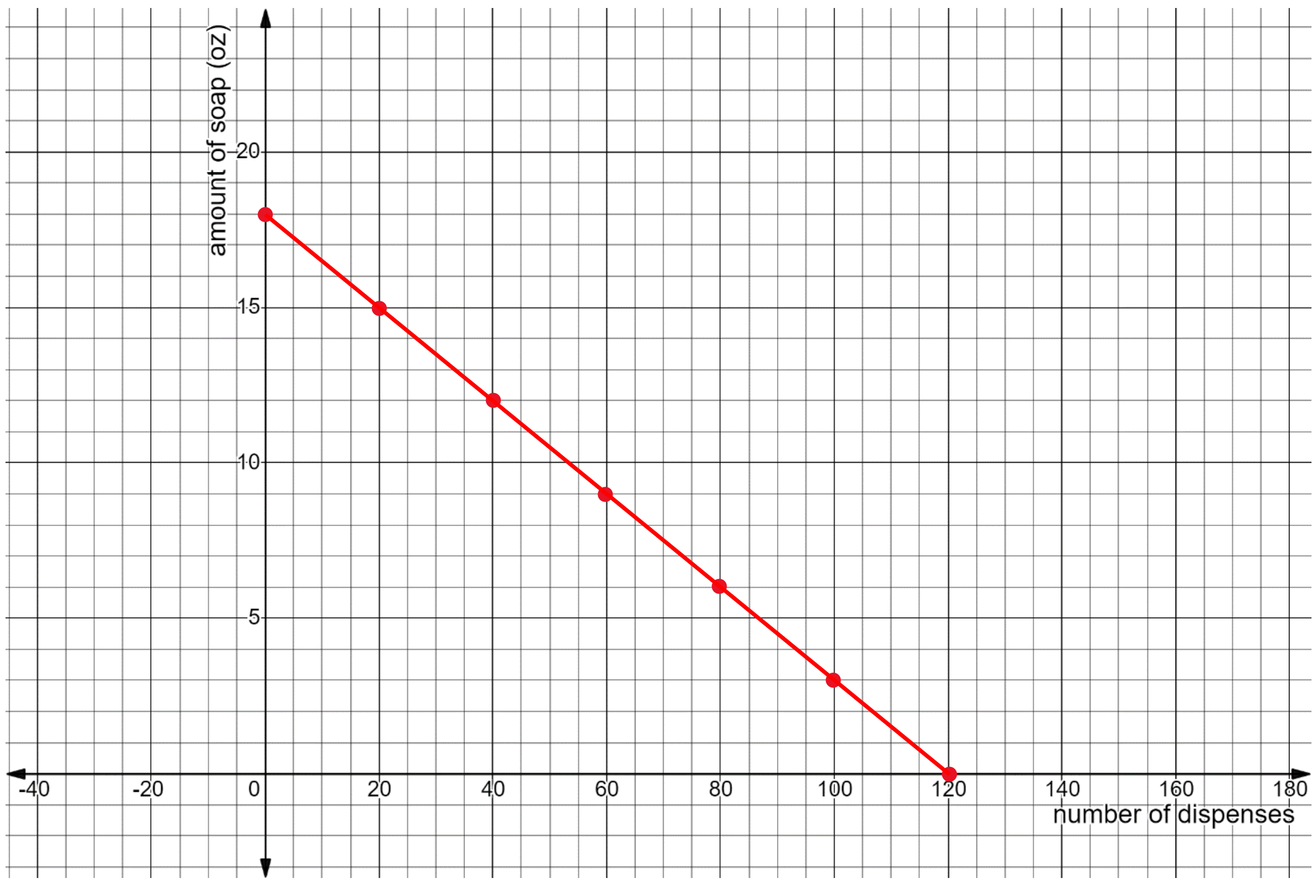
8. What key feature is $f(0)$?

This is the $y$-intercept of the graph. This is the amount of soap when no dispenses are made.

9. Select all of the constraints for the context.

- ☑ $x \geq 0$
- ☐ $x \leq 18$
- ☑ $x \leq 120$
- ☑ $f \geq 0$
- ☑ $f \leq 18$
- ☐ $f \leq 120$

10. Graph the function in terms of the context.



Algebra 1
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Lesson 7 Practice

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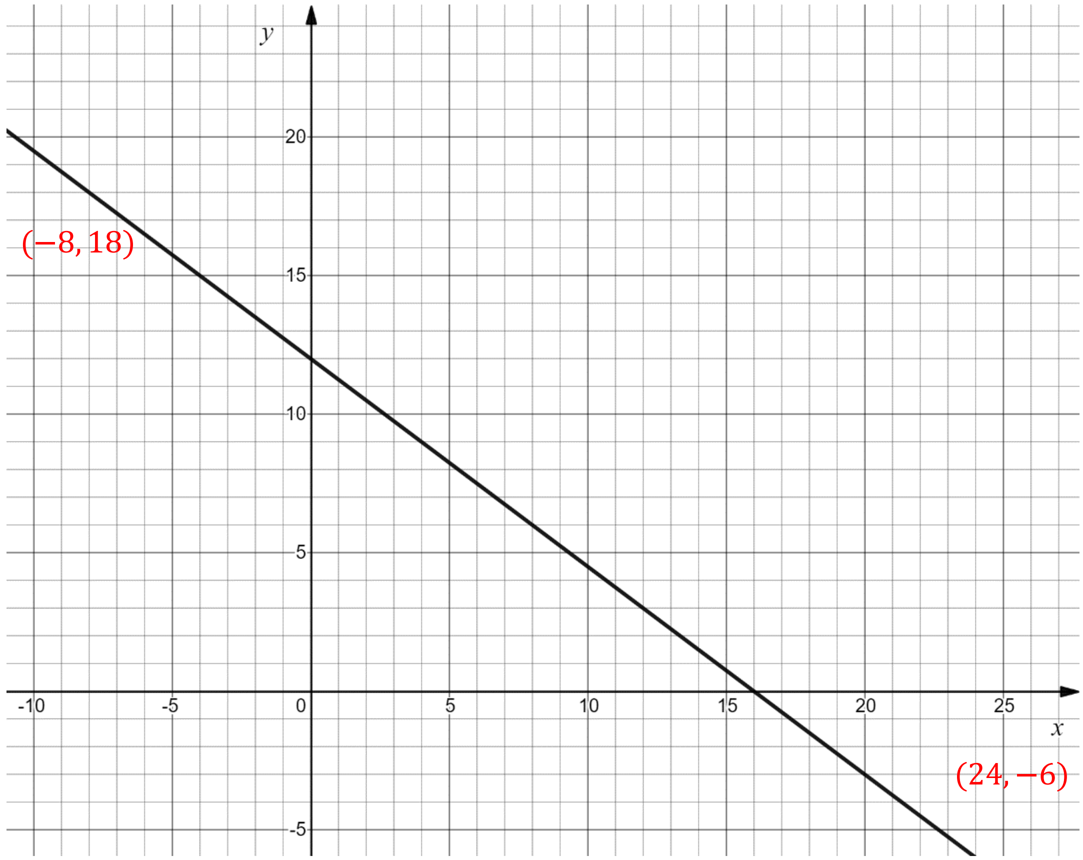
The function $f(x) = -\frac{3}{4}x + 12$ is shown on the coordinate grid.




1. Use the graph to find the solution $f(x) = 18$. Label the point that represents the solution.
   $(-8, 18)$

2. Solve $-\frac{3}{4}x + 12 = 18$. Show your work.

   $-\frac{3}{4}x + 12 = 18$

   $-12 \quad -12$

   $-\frac{4}{3} \cdot -\frac{3}{4}x = 6 \cdot -\frac{4}{3}$

   $x = -\frac{24}{3}$

   $x = -8$

3. Use the graph to find the solution $f(x) = -6$. Label the point that represents the solution.
$(24, -6)$

4. Solve $-\frac{3}{4}x + 12 = -6$. Show your work.
$-\frac{3}{4}x + 12 = -6$
$-12 \quad -12$
$-\frac{4}{3} \cdot -\frac{3}{4}x = -18 \cdot -\frac{4}{3}$
$x = \frac{72}{3}$
$x = 24$

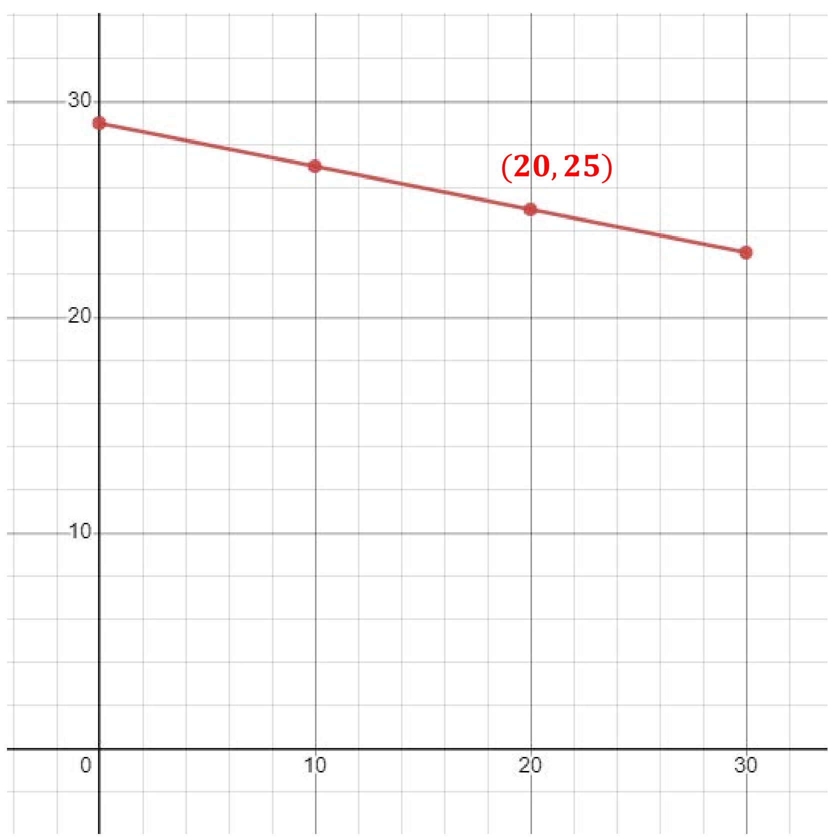
From 1990 to 2020, in Nigeria the percentage of land that are forests has declined. The function $F(t) = -0.2t + 29$ models the percentage of the land, $F$, that is forested where $t$ is the number of years since 1990.

5. Explain why the values of $t$ should have a constraint.
$t$ should have a constraint because the equation only applies to the years between 1990 and 2020. There can't be negative years since 1990.

6. Write the constraint for the values of $t$ using set builder notation.
$\{t | 0 \leq t \leq 30\}$

7. Write the constraint for the values of $F$ using set builder notation.
$\{F | 23 \leq F \leq 29\}$

8. Graph the function using the constraints.



9. Use the graph to find the solution $F(t) = 25$. Label the point that represents the solution.

$25 = -0.2t + 29$

$-29 \qquad\qquad -29$

$\frac{-4}{-0.2} = -\frac{0.2t}{-0.2}$

$t = 20$

10. Solve $0.5t + 47 = 55$. Show your work.

$-47 \quad -47$

$\frac{0.5t}{0.5} = \frac{8}{0.5}$

$t = 16$

Algebra 1
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Lesson 8 Practice

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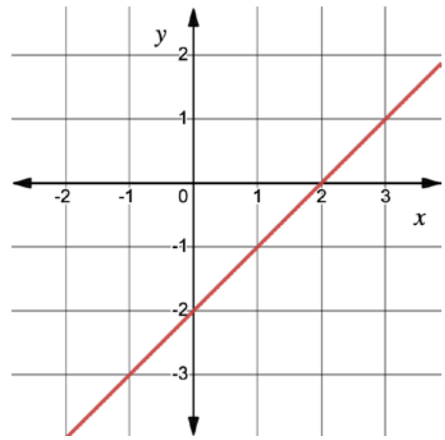
For questions 1 through 4, use the table showing Lines $B$ and $M$.

| **Line $B$** | **Line $M$** |
|---|---|
| $B(x) = \frac{3}{5}x - 4$ |  |

1. Which linear function has the steeper slope?
   Line $M$

2. Which line has the larger $x$-intercept?
   Line $B$

3. Name all the key feature(s) that are the same for both linear functions.
   Both linear functions have the same domain and range. They are also both increasing.

4. Create a linear function that has the same $y$-intercept as Line $B$.
   Answers will vary. $C(x) = \frac{1}{4}x - 4$

5. Complete the statements below to describe the end behavior of Line $B$.

   a. As $x \to +\infty$, $y \to$
   o $-\infty$.
   ✓ $\infty$.

   b. As $x \to -\infty$, $y \to$
   ✓ $-\infty$.
   o $\infty$.

For questions 6 through 10, use the table showing Lines $G$ and $R$.

<table>
<tr><th>Line $G$</th><th>Line $R$</th></tr>
<tr>
<td>$x$-int: $-6$<br><br>A line passes through the points $(-6, 0)$ and $(4, 15)$.<br><br>Slope: $\frac{15-0}{4--6} = \frac{15}{10} = \frac{3}{2}$<br><br>$y$-int: $0 = \frac{3}{2}(-6) + b$<br>$0 = -9 + b$<br>$9 = b$</td>
<td>Slope: $\frac{5-4}{0--3} = \frac{1}{3}$     $y$-int: 5<br>
<table>
<tr><th>$x$</th><th>$y$</th></tr>
<tr><td>$-3$</td><td>$4$</td></tr>
<tr><td>$0$</td><td>$5$</td></tr>
<tr><td>$3$</td><td>$6$</td></tr>
<tr><td>$6$</td><td>$7$</td></tr>
</table>
</td>
</tr>
</table>

$x$-int: $0 = \frac{1}{3}x + 5$
$-5 = \frac{1}{3}x$
$-15 = x$

6. Which linear function has the smaller slope?
Line $R$

7. Which linear function has the larger $y$-intercept?
Line $G$

8. Name the key feature(s) that are different for the functions.
Line $G$ and Line $R$ have different $x$- and $y$-intercepts and slopes.

9. Create a linear function that has the same slope as Line $R$.
Answers will vary. $D(x) = \frac{1}{3}x - 6$

10. Consider the end behavior for the two functions.

a. Complete the statement.

Lines $G$ and $R$ have
- ✓ the same end behavior.
- o different end behaviors.

b. Write the end behavior of the functions.
As $x \to \infty$, $y \to \infty$.
As $x \to -\infty$, $y \to -\infty$.

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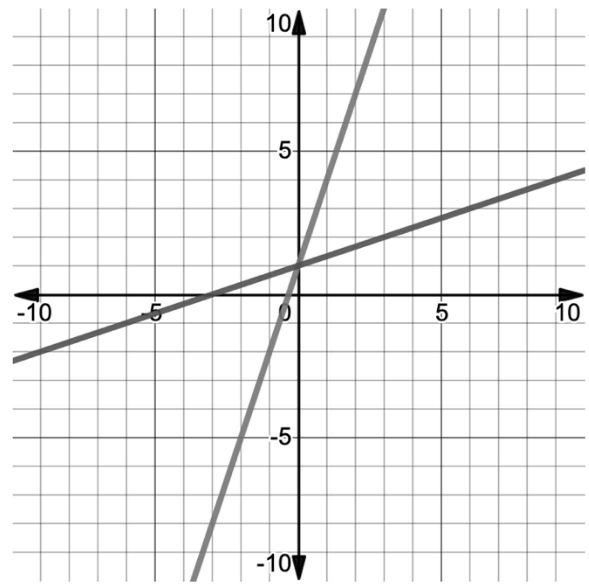
Use the following information to answer questions 1 through 5.

The lines of the equations $y-7=3(x-2)$ and $-\frac{1}{3}x+y=1$ are shown on the same coordinate grid.




1. What is the slope of the line representing the equation $y-7=3(x-2)$?

Slope= $3$

$y-y_1=m(x-x_1)$

$m$ =slope

2. What is the slope of the line representing the equation $-\frac{1}{3}x+y=1$?

Slope= $\frac{1}{3}$

$+\frac{1}{3x}$ $+\frac{1}{3}x$

$y=\frac{1}{3}x+1$

$y=mx+b$

$m$ =slope

3. What is the slope of a line that is perpendicular to $-\frac{1}{3}x+y=1$?

Perpendicular will be opposite sign and reciprocal.

$y=\frac{1}{3}x+1$

$\frac{1}{3} \perp -3$

$=-3$

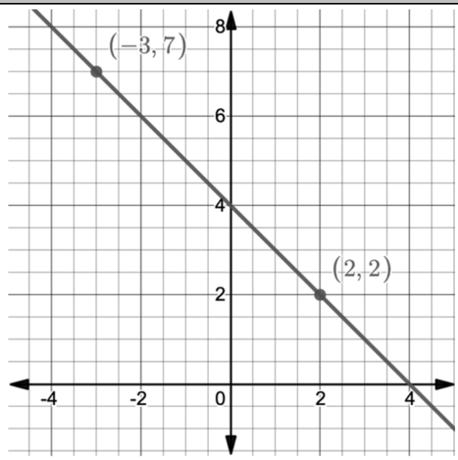
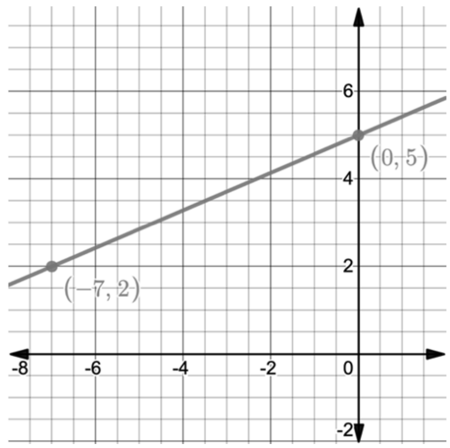
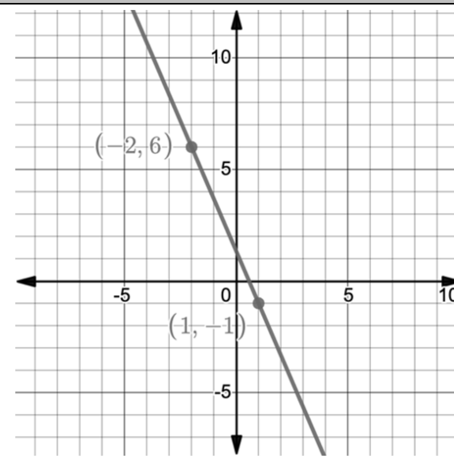
4. Explain how to determine if two lines are perpendicular.

Find their slopes and compare them. They should be opposites and reciprocals.

5. Determine if the lines are parallel, perpendicular, or neither.

The lines are neither. $\frac{1}{3}$ vs $3$ are neither parallel nor perpendicular.

Use the following lines to answer questions 6 through 8.

| **Line $D$** | **Line $G$** | **Line $K$** |
|---|---|---|
|  |  |  |
| **Line $M$** | **Line $N$** | **Line $P$** |
| $-x-y=6$ | $y=-x+8$ | $\frac{7}{3}x-y=-1$ |

6. Which lines are parallel? $\frac{y_2-y_1}{x_2-x_1}$ or $\frac{\text{rise}}{\text{run}}$ parallel = same slope

Slopes: $D: -1$ $M: -1$
$G: \frac{3}{7}$ $N: -1$
$K: \frac{-7}{3}$ $P: \frac{7}{3}$

$-x-y=6$
$-y=x+6$
$y=-x-6$

$\frac{7}{3}x-y=-1$
$-y=\frac{-7}{3}x-1$
$y=\frac{7}{3}x+1$

Same slope = $D, M$ and $N$

7. Which lines are perpendicular?
$\frac{3}{7}$ and $-\frac{7}{3}$ opposite sign reciprocals = $G$ and $K$

8. What is the slope of a line that is parallel to Line $P$?
Line $P = \frac{7}{3}$ Parallel to $P = \frac{7}{3}$

9. Determine if the lines of the two equations are parallel, perpendicular, or neither.

| **Equation $H$** | **Equation $D$** |
|---|---|
| $-2x-4y=-8$ | $-2x+4y=-8$ |

$H: -4y = 2x - 8$
$y = -\frac{2}{4}x + 2$
$m = \frac{-1}{2}$

$D: \frac{4y}{4} = \frac{2x-8}{4}$
$y = \frac{1}{2}x - 2$
$m = \frac{1}{2}$

$-\frac{1}{2}$ vs $\frac{1}{2}$ = neither

10. Determine if the lines of the two equations are parallel, perpendicular, or neither.

| **Equation $S$** | **Equation $W$** |
|---|---|
| $-5x + y = -3$ | $y - 8 = 5(x - 3)$ |

$S$: $y = 5x - 3$ $\qquad$ $W$: $m = 5$

$m = 5$

Same slope = parallel

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Lesson 10 Practice

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1. Write the equation of the line that passes through $(\overset{x}{-2}, \overset{y}{1})$ and is parallel to $y = 2x - 3$ in point-slope form.

$m = 2$

$y - y_1 = m(x - x_1)$

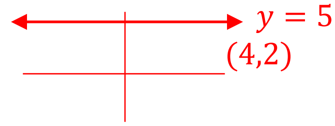
$\boxed{y - 1 = 2(x + 2)}$

2. Write the equation of the line that is parallel to $y = 5$ and passes through point $(4, 2)$.

horizontal line

$m = 0$

$\boxed{y = 2}$




3. Write the equation of the line that contains point $(\underset{x}{0}, \underset{y}{-3})$ and is parallel to $4x + 3y = 6$ in slope-intercept form.

$-4x \qquad -4x$

$\frac{3y}{3} = \frac{6}{3} - \frac{4x}{3}$

$y = 2 - \frac{4}{3x}$

$m = -\frac{4}{3}$

$(0, -3) \downarrow$ $y$-intercept

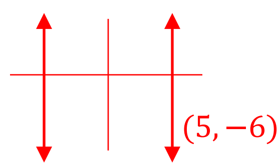
$\boxed{y = -\frac{4}{3}x - 3}$

4. Write the equation of the line that is parallel to $x + 6 = 0$ and passes through point $(5, -6)$.

$\underline{\quad -6 \quad -6}$

$x = -6$ (vertical line)

$\boxed{x = 5}$




5. Write the equation of the line that passes through point $(3, -4)$ and is parallel to $y + \frac{1}{7} = -\frac{8}{7}(x - 1)$ in standard form.

$-\frac{8}{7}x$

Slope$= m$

$-\frac{8}{7}$

$y + 4 = -\frac{8}{7}(x - 3)$

$y(7) + 4(7) = \frac{-8(7)}{7}x + \frac{24(7)}{7}$

$7y + 28 = -8x + 24$

$+8x \qquad\qquad +8x$

$7y + 8x + 28 = 24$

$\qquad -28 \quad -28$

$\boxed{8x + 7y = -4}$

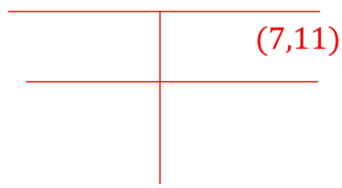
6. Write the equation of the line, in all three forms, that passes through point $\underset{x}{(-4},\underset{y}{11)}$ and is parallel to $y+8=-\frac{5}{3}(x-3)$.   $-\frac{5}{3}=$ slope

| Point-Slope Form | Slope-Intercept Form | Standard Form |
|---|---|---|
| $y-11=-\frac{5}{3}(x+4)$ | $y-11=-\frac{5}{3}x-\frac{20}{3}$<br>$+11 \qquad\qquad + 11$<br>$y=-\frac{5}{3}x+\frac{13}{3}$ | $(3)y=-\frac{5(3)}{3}x+\frac{13}{3}(3)$<br>$3y=-5x+13$<br>$\underline{+5x \quad +5x}$<br>$5x+3y=13$ |

7. Write the equation of a line that is parallel to $y=0$ and contains point $(7,11)$.

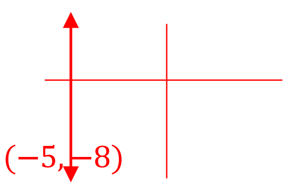
horizontal

$\boxed{y=11}$




$x=$ vertical

8. Line $K$ passes through $(-5,-8)$ and is parallel Line $R$ which has an undefined slope. Write the equation of Line $K$.




$\boxed{x=-5}$

9. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(10,1)$ and $(6,-1)$ and passes through point $(8,3)$.

| $x$ | $y$ |
|---|---|
| 10 | 1 |
| 6 | −1 |

$-4 \qquad -2$

$\frac{\Delta y}{\Delta x}=\frac{-2}{-4}=\frac{1}{2}$

| Point-Slope Form | Slope-Intercept Form | Standard Form |
|---|---|---|
| $\boxed{y-3=\frac{1}{2}(x-8)}$ | $y-3=\frac{1}{2}(x-8)$<br>$y-3=\frac{1}{2}x-4$<br>$\underline{+3 \qquad\quad +3}$<br>$\boxed{y=\frac{1}{2}x-1}$ | $(2)y=\frac{1(2)}{2}x-1(2)$<br>$2y=x-2$<br>$\underline{-x \quad -x}$<br>$\boxed{-x+2y=-2}$ |

10. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(2, 7)$ and $(-1, 17)$ and passes through point $(2, 20)$.

$$\begin{array}{c|c} x & y \\ \hline 2 & 7 \\ -1 & 17 \end{array}$$

$-3$ $+10$

$m = \frac{\Delta y}{\Delta x} = \frac{10}{-3}$

| Point-Slope Form | Slope-Intercept Form | Standard Form |
|---|---|---|
| $\boxed{y - 20 = -\frac{10}{3}(x - 2)}$ | $y - 20 = \frac{-10}{3}(x - 2)$<br>$y - 20 = \frac{-10}{3}x + \frac{20}{3}$<br>$\boxed{y = -\frac{10}{3}x + \frac{80}{3}}$ | $(3)y = \frac{-10(3)}{3}x + \frac{80(3)}{3}$<br>$3y = -10x + 80$<br>$\underline{+10x \ \ +10x}$<br>$\boxed{10x + 3y = 80}$ |

Algebra 1
Unit 4
Lesson 11 Practice

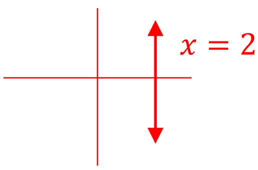
Name Answer Key
Date
Period

1. Write the equation of the line that passes through $\overset{x,y}{(4,2)}$ and is perpendicular to $y = 3x + 1$ in point-slope form.

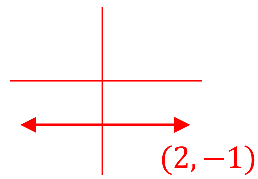
$m = -\frac{1}{3}$ $\qquad$ $y - 2 = -\frac{1}{3}(x - 4)$

2. Write the equation in slope-intercept form of the line that is perpendicular to $x = 2$ and passes through point $(2, -1)$.

Vertical line $\qquad$ Horizontal line




$\boxed{y = -1}$




3. Write the equation of the line that contains point $\overset{x\ y}{(-5, 5)}$ and is perpendicular to $\frac{5}{9}x - y = 4$ in standard form.

$y = \frac{5}{9}x - 4$

$\perp m = -\frac{9}{5}$

$y - 5 = \frac{-9}{5}(x + 5)$

$y - 5 = \frac{-9}{5}x - 9$

$\quad +9 \qquad\qquad +9$

$y + 4 = -\frac{9}{5}x$

$(5)y + \frac{9(5)}{5}x = -4(5)$

$\boxed{9x + 5y = -20}$

4. Write the equation of the line that is perpendicular to $2x + 5y = 6$ and passes through the origin. $(0,0)$

$\perp m = \frac{5}{2}$ $\quad (0,0)$

$\boxed{y = \frac{5}{2}x}$

$\frac{5y}{5} = \frac{6-2x}{5}$

$y = -\frac{2}{5}x + \frac{6}{5}$

5. Line $F$ passes through $(-12, 0)$ and is perpendicular to Line $Z$ which has an undefined slope. What is the slope of line $F$?

$x =$ line

Slope of line $F$ is $0$ since it is a horizontal line.

6. Write the equation of the line that is perpendicular to the line that passes through the points $(7, -3)$ and $(7, 14)$ and passes through point $\overset{x\ y}{\left(4, -\frac{1}{2}\right)}$.

| | $x$ | $y$ | |
|---|---|---|---|
| 0 | 7 | $-3$ | +17 |
| | 7 | 14 | |

$m = \frac{17}{0}$

$y = -\frac{1}{2}$

Undefined → vertical line – perpendicular is a horizontal line.

7. Write the equation of a line that is perpendicular to $-y = -x$ and contains point $(-1, -1)$.

$\frac{-y}{-1} = \frac{-x}{-1}$

$y = x \qquad m = 1$

Slope of $\perp$ line $= -1$

$\boxed{y + 1 = -(x + 1) \text{ or } y = -x -}$

8. Write the equation of the line, in all three forms, that passes through point $(6,-2)$ and is perpendicular to $y+3=-7(x+1)$. $m=-7$ $\perp$ slope $=\frac{1}{7}$

| Point-Slope Form | Slope-Intercept Form | Standard Form |
|---|---|---|
| $y+2=\frac{1}{7}(x-6)$ | $y+2=\frac{1}{7}x-\frac{6}{7}$<br>$\underline{-2\quad\quad -2}$<br>$\boxed{y=\frac{1}{7}x-\frac{20}{7}}$ | $(7)y=\frac{1(7)}{7}x-\frac{20}{7}(7)$<br>$7y=x-20$<br>$\underline{-x\quad -x}$<br>$\boxed{-x+7y=-20}$ *or*<br>x-7y=20 |

9. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(20,0)$ and $(0,-10)$ and passes through point $(-10,10)$.

$x$ $y$

| $x$ | $y$ |
|---|---|
| 20 | 0 |
| 0 | −10 |

−20 −10

$m=\frac{-10}{-20}=\frac{1}{2}$

$\perp m=-2$

| Point-Slope Form | Slope-Intercept Form | Standard Form |
|---|---|---|
| $y-10=-2(x+10)$ | $y-10$<br>$=-2x-20$<br>$\boxed{+10}+10$<br>$y=-2x-10$ | $2x+y=-10$ |

10. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $\underline{(8,9)\text{ and }(-4,-7)}$ and passes through point $(0,-5)$.

$x$ $y$

| $x$ | $y$ |
|---|---|
| 8 | 9 |
| −4 | −7 |

−12 −16

| Point-Slope Form | Slope-Intercept Form | Standard Form |
|---|---|---|
| $y+5=-\frac{3}{4}(x-0)$<br>OR<br>$y+5=-\frac{3}{4}x$ | $y=-\frac{3}{4}x-5$ | $(4)y=\frac{-3(4)}{4}x-5(4)$<br>$4y=-3x-20$<br>$\underline{+3x\quad +3x}$<br>$3x+4y=-20$ |

$\frac{\Delta y}{\Delta x}=\frac{-16}{-12}=\frac{8}{6}=\frac{4}{3}$

So $\perp m=-\frac{3}{4}$

Algebra 1 | Name Answer Key
Unit 4 | Date
Lesson 12 Practice | Period

1. Write the equation of the line that passes through $(3, 2)$ and is parallel to $3x - 2y = 1$.

$-3x \quad -3x$

$\frac{-2y}{-2} = \frac{1}{-2} - \frac{3x}{-2}$

$y = \frac{3}{2}x - \frac{1}{2}$

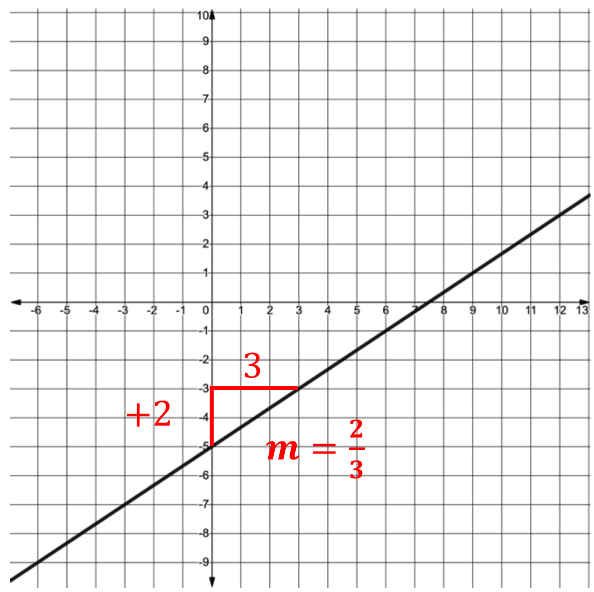
$m = \frac{3}{2}$ $\quad \overset{x\ y}{(3,2)}$

$\boxed{y - 2 = \frac{3}{2}(x - 3)}$ OR $\boxed{y = \frac{3}{2}x - \frac{5}{2}}$

$y - 2 = \frac{3}{2}x - \frac{9}{2}$

$+2 \qquad +2$

Use the line shown on the coordinate grid to complete questions 2 through 4.




2. What is the slope of the line that is perpendicular to the line shown on the coordinate grid?

$m = -\frac{3}{2}$

3. Write the equation of the line that is perpendicular to the line shown on the coordinate grid and that passes through $\overset{}{(1, 0)}$.
$x\ y$

$y - 0 = \frac{-3}{2}(x - 1)$

$\boxed{y = -\frac{3}{2}x + \frac{3}{2}}$

4. Write the equation of the line that is parallel to the line shown on the coordinate grid and that passes through $(5, -4)$

$m = \frac{2}{3}$

$\boxed{y + 4 = \frac{2}{3}(x - 5) \text{ or } y = \frac{2}{3}x - \frac{22}{3}}$

5. Write the equation of the line in standard form that is perpendicular to $y = -\frac{1}{4}x - 13$ and passes through $(1, 7)$.

$\perp m = 4$

$y - 7 = 4(x - 1)$

$y - 7 = 4x - 4$

$\underline{-4x \quad -4x}$

$y - 4x - 7 = -4$

$+7 \quad +7$

$\boxed{-4x + y = 3 \text{ or } 4x - y = -3}$

Use the following information about Line $U$ to complete questions 6 through 8.

Line $U$ passes through $(-2,-2.5)$ and $(1,0)$.

6. Line $Z$ is parallel to Line $U$. What is the slope of Line $Z$?

| $x$ | $y$ |
|---|---|
| $-2$ | $-2.5$ |
| $1$ | $0$ |

$+3$ $\qquad$ $+2.5$

$$\frac{\Delta y}{\Delta x} = \frac{2.5}{3} = \frac{5}{6}$$

$\boxed{\text{Slope of line } Z = \frac{5}{6}}$

7. Line $Z$ passes through $(\underset{x}{-6},\underset{y}{9})$. Write the equation of Line $Z$ in slope-intercept form.

$y - 9 = \frac{5}{6}(x+6)$

$y - 9 = \frac{5}{6}x + 5$

$+9 \qquad +9$

$\boxed{y = \frac{5}{6}x + 14}$

8. Line $C$ passes through $(\overset{x}{7},\overset{y}{3})$ and is perpendicular to Line $U$. Write the equation of Line $C$ in point-slope and standard forms. $\perp m = -\frac{6}{5}$

| Point-Slope Form | Standard Form |
|---|---|
| $\boxed{y - 3 = -\frac{6}{5}(x-7)}$ | $(5)y - 3(5) = \frac{-6(5)}{5}x + \frac{42(5)}{5}$<br>$5y - 15 = -6x + 42$<br>$\underline{+6x + 15 \quad + 6x + 15}$<br>$\boxed{6x + 5y = 57}$ |

9. Write the equation of the line that is parallel to $-21x + 18y = 3$ and passes through point $(\underset{x}{-6},\underset{y}{-3})$ in point-slope form.

$\underline{+21x \qquad\qquad +21x}$

$\frac{18y}{18} = \frac{3}{18} + \frac{21x}{18}$

$y = \frac{7}{6}x + \frac{1}{6}$

$m = \frac{7}{6}$

$\boxed{y + 3 = \frac{7}{6}(x+6)}$

10. Write the equation of the line that is perpendicular to $10x = -y$ and passes through point $\left(\underset{x}{4},\underset{y}{\frac{2}{5}}\right)$.

$\underline{10x} = \underline{-y}$
$-1 \qquad -1$

$y = -10x$

$\perp m = \frac{1}{10}$

$\boxed{y - \frac{2}{5} = \frac{1}{10}(x-4)}$